Theory paper — answer sheets (final results only)

Full working is in the full working (working.pdf); every number is checked by `python theory/check_numbers.py`.

T1

- **A.1 (iii)** q_1 = "How does weight-decay strength affect the generalisation gap of transformer language models across sizes from 10M to 10B parameters?" $\succeq$ q_2 = the same question for a 125M-parameter model. **(iv)** For existential "Can X improve Y?" questions one witness for the narrow question also answers the broad one, so the narrower question subsumes the broader one, not the reverse.
- **A.2** Angles $0^\circ, 30^\circ, 60^\circ$: $u = (1, 0)$, $v = (0.866, 0.5)$, $w = (0.5, 0.866)$; $\cos(u, v) = 0.866$, $\cos(v, w) = 0.866$, $\cos(u, w) = 0.500$.
- **B.1** Invisible: role swap (and any near-miss made purely by reordering tokens). Visible: constraint flip, goal swap, inquiry change, direction flip.
- **C.1** Control set: length-flipped scope pairs (broader question is the longer one). Statistic: scope-order accuracy on those pairs (≈0 % for a length model).
- **C.2** (ii) Direction decided by: the volume ratio $\mathrm{Vol}(A)/\mathrm{Vol}(B)$ alone. (iii) $A = [0, 10] \times [0, 1]$, $B = [9.9, 10.9] \times [0, 1]$: $\mathrm{Vol}(A) = 10\,\mathrm{Vol}(B)$, $P(B \subseteq A) = 0.1$.

T2

- **A.1** G1 How far is x from the closest thing already written? (1, 9) · G2 Is x in a sparsely populated region of a reference population? (2, 6) · G3 Does x reach outside or join what the field spans? (3, 4) · G4 Has exactly this been done before? (5) · G5 Would a reader call it new? (8) · G6 Did the field later go where x went? (7) · G7 Did independent people build on x? (10)
- **B.1** $Y(j) = 1 - 1/\sqrt{m}$, $N(j) = 1 - j/m$; satisfies the gradient axiom: N.
- **B.2** $Y = 1 - \sqrt{j/m}$, $N = 1 - \sqrt{j}/m$.
- **B.3** Satisfies: Y. Violates: N.
- **B.4** $M = 1 - \frac{1}{2}\left[\frac{1}{m}\sum_i \max_{p \in P} \cos^+(c_i, p) + \frac{1}{m}\max_{p \in P}\left(\sum_i \cos^+(c_i, p)\right)^2\right]$
- **C.1** Reliability $= 0.529$. Largest correlation $= 0.727$.
- **C.2** Partial correlation $= 0.181$.

T3

- **A** (1) Evaluate on length-matched pairs. (2) Control length statistically (length-only baseline / covariate).
- **B** Over all battles $= 0.68$. Within systems $= 0.50$.
- **C(i)** Share already asked $= 0.915$.

- **C(ii)** G1: 100 % already asked → 10.0 % realised later, 0 net-new. G2: 5 % already asked → 0.88 % realised later. G1 tops the leaderboard.
- **D** (1) Rewrite both sides through the same model. (2) Same-author-type pairs with quality labels (human–human, model–model). (3) Swap model families and capability levels. (4) Minimal model polish of human questions.
- **E(ii)** Pairs needed $= 1225$. Half-width with 400 pairs $= 0.049$.

T4

- **A.1** Expected in-plane share $= (k-1)/d = 29/768 = 0.0378$.
- **B** Estimated true angle $\approx 40.9°$.
- **C** (1) Writing quality/explicitness → match on length and extraction self-agreement; rewrite all questions through one model. (2) Reference-list density → subsample R to fixed k with abstracts; match on count and recency. (3) Topic mix → within-topic comparisons.
- **D(ii)** Count proximity only when q is not already asked, not answered by R, and has an off-plane residual above the decoy baseline.

T5

- **A.2** Q_1: $v = 0.531$, $H(Y) = 1.000$ · Q_2: $v = 0.000$, $H(Y) = 1.000$ · Q_3: $v = 0.025$, $H(Y) = 0.494$
- **B.1** Q_{near}: "Does increasing augmentation strength lower the optimal batch size in contrastive pre-training?" ($\pi = 0.5, a = 0.9, b = 0.1, v = 0.531$). Q_{far}: "Does the order in which GPU nodes are allocated change the final accuracy of contrastive pre-training?" ($\pi = 0.5, a = b = 0.6$, $v = 0$). Distance ranks Q_{far} higher; value ranks Q_{near} higher.
- **B.2** (1) Reports-hidden control. (2) Identical question text with reports that make it open vs settled. (3) Test on a measurement family never seen in training.